



Fig. 6: Robotic avatar (Credit : <https://global.toyota>)



Fig. 7: Digit delivery robot (Credit: www.agilityrobotics.com)

2017 by Russia and Japan, respectively. FEDOR avatar robot is able to replicate the movements of a remote operator and perform some actions autonomously. It was sent on an experimental mission to the International Space Station in 2019.

T-HR3 is the third generation humanoid robot from Toyota, Japan. It is controlled from a

master maneuvering system that allows the entire body of the robot to be operated remotely. The master system may include wearable devices to control movements of the robot and a head-mounted display that allows the operator to see from the robot's perspective. That is, it can mimic the movements of its human operator like a real-world avatar.

In future, these humanoids will be able to perform surgeries while operators (human doctors) will be controlling them from another part of the world. These can also be used by caregivers to work with patients remotely. Avatar robots could change the way people interact and work.

6. First delivery humanoid robot

A two-legged robot called Digit (Fig. 7) from Agility Robotics was announced in February 2019. It is

approximately the size and shape of a small adult human. It has advanced feet that allow it to balance on one foot, or carefully navigate through obstacles with both legs. It can navigate environments semi-autonomously with the help of LiDAR and other sensors, enabling it to perceive and map the world for navigation. It can carry boxes up to 18kg in weight in its arms. It can be used in many areas, including logistics, warehouses, telepresence, and industrial inspection.

7. First virtual artificial humans

Neon (Fig. 8), a humanoid artificial intelligence (AI) avatar designed by Samsung's STAR Labs, looks like a human and can respond to questions. It is basically a next-generation AI video chatbot with a human face and personality but not exactly an AI assistant. You can imagine something like Siri or Alexa with a virtual face that could start a conversation. It interacts with humans along with expressions such as a smile or a raised eyebrow. It is still under development but could be the next big thing in the future for use in many fields, including retail, tourism, collaboration, healthcare, among others.

8. First robonaut

A robonaut (Fig. 9) is a humanoid robot for the astronauts. The first-generation robonaut was designed by the Robot Systems Technology Branch at NASA's Johnson Space Center. The core idea behind the robonaut is to have a humanoid machine work alongside astronauts. Many countries are working on their own robonauts



Fig. 8: Neon-virtual humanoid (Credit: www.cnet.com)



Fig. 9: Robonauts (Credit: www.wikipedia.org)